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Trans-generational effects in the clonal invader *Alternanthera philoxeroides*

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Abstract

Aims Recent studies have revealed heritable phenotypic plasticity through vegetative generations. In this sense, changes in gene regulation induced by the environment, such as DNA methylation (i.e. epigenetic changes), can result in reversible plastic responses being transferred to the offspring generations. This trans-generational plasticity is expected to be especially relevant in clonal plants, since reduction of sexual reproduction can decrease the potential for adaptation through genetic variation. Many of the most aggressive plant invaders are clonal, and clonality has been suggested as key to explain plant invasiveness. Here we aim to determine whether trans-generational effects occur in the clonal invader *Alternanthera philoxeroides*, and whether such effects differ between populations from native and non-native ranges.

Methods In a common garden experiment, parent plants of *A. philoxeroides* from populations collected in Brazil (native range) and Iberian Peninsula (non-native range) were grown in high and low soil nutrient conditions, and offspring plants were transplanted to control conditions with high nutrients. To test the potential role of DNA methylation on trans-generational plasticity, half of the parent plants were treated with the demethylating agent, 5-azacytidine.

Important Findings Trans-generational effects were observed both in populations from the native and the non-native ranges. Interestingly, trans-generational effects occurred on growth variables (number of ramets, stem mass, root mass and total mass) in the population from the native range, but on biomass partitioning in the population from the non-native range. Trans-generational effects of the population from the native range may be explained by a 'silver-spoon' effect, whereas those of the population from the non-native range could be explained by epigenetic transmission due to DNA methylation. Our study highlights the importance of trans-generational effects on the growth of a clonal plant, which could help to understand the mechanisms underlying expansion success of many clonal plants.

Keywords: 5-azacytidine, alligator weed, clonal growth, DNA methylation, epigenetic variation, plant invasions

摘要：近年的研究已经揭示出表型可塑性可以通过营养世代遗传。从这个意义上讲，环境诱导的基因调控变化(例如DNA甲基化，即表观遗传变异)会导致可逆的塑性反应传递给后代。这种跨世代的可塑性在克隆植物中尤其重要，因为有性繁殖减少会降低通过遗传变异进行适应的可能性。许多最具侵略性的植物入侵者都具有克隆性，而且克隆性被认为是植物入侵性的关键。本研究中，我们的目标是确定在克隆入侵者喜旱莲子草(*Alternanthera philoxeroides*)中是否发生了跨代效应，以及这种效应在本地和非本地群落之间是否存在差异。在同质园实验中，我们将采自巴西(原产地)和伊比利亚半岛(非原产地)的喜旱莲子草种群分别种植在养分高和低的土壤中，后代植株被移植到高养分条件下作为对照。为了检验DNA甲基化对跨代可塑性的潜在作用，一半的母株用去甲基化剂5-氮杂胞苷处理。在来自原产地和非原产地的种群中均观察到了跨代效应。有趣的是，在原产地种群的生长变量(分株数、茎生物量、根生物量和总生物量)中发现了跨代效应，而

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在非原产地种群中发现了生物量分配具有跨代效应。在原产地种群中出现的跨代效应可以用“银勺”效应来解释，而非原产地种群的跨代效应可以归因于DNA甲基化引起的表观遗传传递。本研究强调了跨代效应对克隆植物生长的重要性，这有助于理解众多克隆植物能够成功繁殖扩张的机制。

关键词：5-氮杂胞苷、喜旱莲子草、克隆生长、DNA 甲基化、表观遗传变异、植物入侵

INTRODUCTION

Invasive species represent a significant threat to global biodiversity, decreasing the abundance and richness of native species (Gaertner *et al.* 2009; McGeoch *et al.* 2010). Discovering the underlying mechanisms of biological invasions is a fast-moving research topic in modern ecology. One core question is to determine why some species have the potential to become invasive when naturalized outside their native range (Alpert *et al.* 2000). A widely used approach is to study those life-history traits of the invasive species that allow or facilitate this invasion process (Hamilton *et al.* 2005; Kolar and Lodge 2001).

Phenotypic plasticity is the ability of an organism to respond to different environments by expressing its phenotype according to the influence of the environment in which the organism develops (Bradshaw 1965; Sultan 2000). Plasticity allows a genotype to have a wider tolerance to environmental conditions, and therefore a higher fitness across multiple habitats (Caño *et al.* 2008; Ghalambor *et al.* 2007; van Kleunen and Fischer 2005). This feature is considered relevant in biological invasions, since those species that are more plastic have greater facility to successfully adapt themselves to new environments (Hulme 2008; Smith 2009; Zenni *et al.* 2014). A widely described case of phenotypic plasticity is the changes that occur in response to scarcity of an essential resource for the development of the plant, mainly water, light or nutrients. In such a situation, plants may respond by increasing the allocation of biomass to the structures responsible for obtaining the most limiting resource for growth (Chapin *et al.* 1987; Ryser and Eek 2000). For instance, plasticity in shoot to root ratio allows the plant to maintain an optimal development in conditions of scarcity (Aikio and Markkola 2002; Grossman and Rice 2012).

Interestingly, the environmental conditions experienced by the parental generation may influence the offspring phenotype due to trans-generational effects (Galloway 2010; Galloway and Etterson 2007; Latzel and Klimešová 2010; Sultan *et al.* 2009). For example, resource levels experienced by the parental generation could regulate the phenotype expressed by the offspring generation. Under this premise, it seems logical to anticipate that offspring generations could gain a benefit when occupying an area with similar conditions to those experienced by their parents (Dong *et al.* 2017, 2018). In this sense, trans-generational effects could favour the successful establishment of offspring generations, and therefore be considered adaptive (Galloway and Etterson 2007; Latzel *et al.* 2014). This trans-generational plasticity is expected to be especially relevant in clonal plants, for which reduction of sexual reproduction can decrease the potential for adaptation through genetic variation, and could explain the success of clonal species in many communities (Latzel and Klimešová 2010). Many studies have been conducted to explore trans-generational plasticity in plants (Dong *et al.* 2017, 2018; González *et al.* 2016, 2017; Latzel *et al.* 2010, 2014; Sultan *et al.* 2009). To date, however, less effort has been dedicated to determining the importance of trans-generational plasticity in plant invasiveness (Caño *et al.* 2016; Richards *et al.* 2012).

A number of studies point out that the influence of environmental conditions on phenotypes could be mediated by heritable epigenetic variation (Hallgrímsson and Hall 2011; Verhoeven and Preite 2014), through which trans-generational plasticity can occur (Boyko *et al.* 2010). Heritable epigenetic variation does not involve alterations of the DNA sequence, but involves regulatory mechanisms of gene expression, including the repression or silencing of particular genes (Wolffe and Matzke 1999). This could provide an alternative way for a species to adapt to a new environment, i.e. as opposed to selection for a different genotype (Bossdorf *et al.* 2008). Epigenetic changes also have the potential advantage of being easily reversible (Bender 2004). In organisms with asexual reproduction, epigenetic changes allow an increase of the phenotypic variability of the populations based on the environment in which they develop, without being necessary an alteration in the genome sequence (Gao *et al.* 2010). Changes in gene expression regulated by epigenetic mechanisms are also called trans-generational memory mechanisms, due to the effects they have on the offspring (González *et al.* 2016; Heard and Martienssen 2014). One of the mechanisms by which these epigenetic changes happen is DNA methylation (Bender 2004). Some epigenetic changes are heritable, so they can be transmitted to offspring (Heard and Martienssen 2014; Martienssen and Colot 2001) and used as a strategy to cope with adverse environmental conditions (Asensi-Fabado *et al.* 2016; Dong *et al.* 2017, 2018; Probst and Mittelsten 2015; Secco *et al.* 2017).

Trans-generational plasticity could be considered a favourable trait, allowing invasive plants, especially those that are clonal, to overcome low genetic variation and to adapt successfully to the new environment in which they develop. In this study we assessed trans-generational plasticity in the clonal invader *Alternanthera philoxeroides* to soil nutrients. Parental generations were grown under high and low nutrient conditions, whereas offspring generations were grown in a common environment with high nutrients. Also, to test the potential role of DNA methylation on trans-generational plasticity, half of the parent plants were treated with the demethylating agent, 5-azacytidine. We hypothesized that environmental conditions experienced by parental generations would affect the growth and biomass allocation exhibited by offspring generation (i.e. trans-generational effect). Specifically, we predict (i) that offspring ramets whose parents grew in high nutrients conditions will show greater growth than ramets whose parents grew in low nutrients conditions; and (ii) that offspring ramets whose parents grew in low nutrients conditions will increase the proportional allocation of biomass to roots in comparison with offspring of parents in high nutrient conditions.

MATERIALS AND METHODS

Study species

Alternanthera philoxeroides (Mart.) Griseb. (Amaranthaceae), commonly known as alligator weed, is a creeping perennial herb native to the Parana River region in South America (Julien *et al.* 1995). It is an amphibious plant that can grow both on land and as floating mats in water courses (Lu and Ding 2010). Sexual reproduction is unusual in its introduced ranges (Buckingham 1996; Lu *et al.* 2013), but clonal reproduction from stem buds is common in both aquatic and terrestrial habitats in the invaded areas (Dong *et al.* 2010, 2019; Julien *et al.* 1995; Sainty *et al.* 1998; Yu *et al.* 2009). In the introduced range *A. philoxeroides* displaces native species, leading to monospecific communities, with the consequent loss of biodiversity (Julien and Bourne 1988; Ma and

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Wang 2005). It can completely cover watercourses and block irrigation systems in riparian crop fields, causing floods (Sainty *et al.* 1998; Stewart 1996; Wang *et al.* 2008). The geographical distribution of the species out of their native range includes the south of North America (Buckingham 1996), Italy, France (Anderson *et al.* 2016), Australia, New Zealand (Julien *et al.* 1995), Indonesia, Thailand and China (Sainty *et al.* 1998). *Alternanthera philoxeroides* is considered one of the worst invasive weeds worldwide because of its rapid growth, resistance to control methods, and ecological and economic impacts. Recently, a naturalized population of *A. philoxeroides* has been described at the Iberian Peninsula (Fisterra Cape, NW Spain) (Romero and Amigo 2015) (see Fig. 1).

Plant material

Plant material used in the experiment was collected from three different locations: one at the introduced area (Iberian Peninsula, in Fisterra: 42°56′10″ N, 9°16′13″ W), and two at the native range (Brazil, in São Carlos: 21°59′1″ S, 47°52′43″ W, and Piracicaba: 22°47′14″ S, 47°38′46″ W). From each location a single genotype was collected and propagated. For the three locations plant material was collected in similar terrestrial habitats. By collecting a single genotype we prevent from genetic differences within populations to interfere with the epigenetic effects on character transmission. Thus, in the absence of genetic variation, differences in gene expression should be caused by an epigenetic effect (Bossdorf *et al.* 2008). In this sense, clonal plants can be considered a suitable model for epigenetic studies. The plants were collected between June and August of 2017 and vegetatively propagated in a greenhouse of the Federal University of São Carlos (Brazil).

Experimental design

In October 2017, clonal fragments comprising the first, second and third ramets from the apices were selected from the propagated plant material of the native (São Carlos and Piracicaba) and non-native (Fisterra) populations. We selected 12 clonal fragments from each population, summing a total of 36 clonal fragments (12 fragments × 3 populations). These ramets, which represent the parental generation, were randomly assigned to factorial nutrient and demethylation treatments, ensuring that all the populations were equally represented in each treatment combination (see Fig. 1). For the nutrient treatment half of the ramets grew in potting compost (high nutrients), and the other half in washed sand (low nutrients). Potting compost used for the high nutrient treatment contained all main nutrients and trace elements, and can be considered as fertile soil providing optimal growth conditions. For the demethylation treatment, half of the ramets were treated with a demethylating agent, 5-azacytidine (Sigma-Aldrich Brasil Ltda, São Paulo, Brazil), which is a substance analogous to cytosine that inhibits DNA methylation in eukaryotes (Čihák 1974). 5-Azacytidine has been previously used to test the role of epigenetic variation in phenotypic plasticity and trans-generational adaptation to stress (Bossdorf *et al.* 2010; González *et al.* 2016). The demethylation treatment was applied in mid-November 2017, with six applications carried out over 2 weeks. Plants were sprayed with a 50 µmol/l solution of 5-azacytidine. The solution was applied early in the morning, when stomata are expected to be open, to ensure absorption by leaves. The 5-azacytidine solution was prepared with a mixture of acetic acid and distilled water at a volume ratio of 1:1. Plants that did not receive the demethylating treatment were sprayed with the same solution without the demethylating agent, in order to homogenize any side effects of acetic acid on the plants. Each nutrient and demethylation treatment was replicated three times (n = 3). Ramets of the parental generation grew for 2 months (October and November 2017).

Afterwards, in December 2017, from each ramet of the parental generation three axillary stems were obtained to create the offspring generation (n = 9), totalling 108 offspring fragments, 36 from each original population. The offspring generation grew under high nutrients and not subjected to demethylation (see Fig. 1). The transplant of the offspring generation was performed in the first week of December 2017, and the plants were maintained for 5 months, until harvest in April 2018. The initial size of the offspring generation was determined as the number of leaves for each clonal fragment. Preliminary analysis showed that the initial size of the offspring generation from the studied

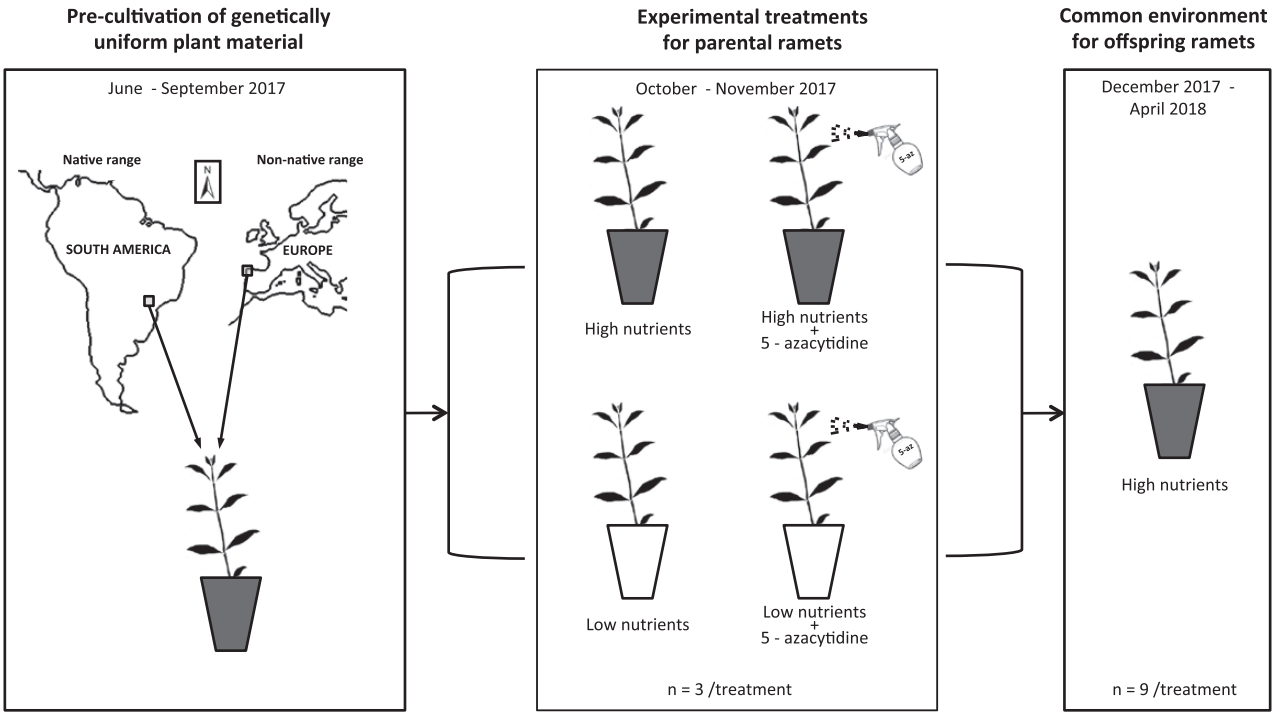

Figure 1: Scheme of the experimental design, including the treatment applied on the parental ramets and the schedule of the experiment.

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populations differed significantly (ANOVA: $F_{2,104}$ = 12.2, P < 0.001), with plants from São Carlos having a higher number of leaves (7.1 ± 2.9, mean ± SE) than plants from Piracicaba (5.4 ± 2.1, mean ± SE) and Fisterra (4.5 ± 2.0, mean ± SE). Parental and offspring ramets were placed in 2.8-l plastic pots, providing enough space to avoid root confining during the experiment. The experiment was carried out in the same greenhouse at the Federal University of São Carlos (Brazil) where plant pre-cultivation was carried out, under a natural day/night light cycle and ambient temperature. Plants were watered regularly to avoid water stress.

Measurements

At the end of the experiment each clonal fragment of the offspring generation was separated into leaves, stems and roots, dried at 60°C for 48 h and weighed. For each clonal fragment we calculated the total mass (leaf mass + stem mass + root mass) and the proportional biomass allocated to roots (root mass ratio: RMR = root mass/total mass). Also, the number of individual ramets in each clonal fragment of the offspring generation was recorded.

Statistical analysis

Analyses were run for the offspring generation. Data were analysed by a three-way ANCOVA with nutrients (high and low nutrients), demethylation (demethylated or not) and population (Fisterra, São Carlos, Piracicaba) as main factors. The initial plant size of the offspring generation, estimated as the number of leaves per clonal fragment, was treated as a covariate to take into account variation among populations. Normality and homoscedasticity were checked using the Kolmogorov–Smirnov and Levene tests. All variables were square-root transformed to fulfil requirements for ANOVA. When results were significant, we applied Tukey tests to detect differences between populations, and between the treatments within each population. Mortality reduced the number of replicates used in the different analyses, as indicated by the error degrees of freedom. All analyses were conducted with the software IBM SPSS Statistics, version 23 (IBM Corp., Armonk, NY).

RESULTS

Nutrient conditions experienced by the parental ramets significantly affected number of ramets, stem mass, root mass and total mass of the offspring generation (Table 1). Thus, when parental ramets grew in high nutrients, their offspring showed significantly greater number of ramets, stem mass, root mass and total mass than offspring whose parents grew under low nutrient conditions (Fig. 2a and c–e). Demethylation treatment significantly reduced leaf mass, independent of the population and of the nutrient conditions experienced by the parental generation (Table 1, Fig. 2b). The interaction between nutrients and demethylation significantly affected root mass and RMR (Table 1). Demethylation increased root mass and RMR in the high nutrient treatment, but reduced them in the low nutrient treatment (Fig. 2d and f).

Population origin significantly affected all the variables except root mass (Table 1). Thus, plants from Fisterra population (non-native range) showed smaller number of ramets, leaf mass, stem mass and total mass than plants from São Carlos and Piracicaba populations (native range) (Fig. 2a–c and e). On the contrary, RMR was significantly greater in Fisterra population (non-native range) than in São Carlos and Piracicaba populations (native range) (Table 1, Fig. 2f). Interestingly, the effect of nutrient status experienced by the parental generation was dependent on the population origin. The interaction between nutrients and population significantly affected the number of ramets, stem mass, total mass and RMR (Table 1). When parent ramets grew in low nutrient conditions, we observed a significant reduction in number of ramets, stem mass and total mass of offspring ramets from São Carlos and Piracicaba populations (native range), but not in those offspring from the Fisterra population (non-native range) (Fig. 2a, c and e). On the contrary, the effect of nutrients on RMR was detected in the Fisterra population (non-native range) but not in the São Carlos and Piracicaba populations (native range). Thus, when parent ramets grew in high nutrient conditions, their offspring significantly reduced RMR, and this effect was only detected in the Fisterra population (non-native range) (Fig. 2f). Remarkably, the Tukey test detected that low nutrients in the parental environment significantly increased RMR of offspring from the Fisterra population (non-native range), but this increase was only detected in plants not demethylated (Fig. 2f).

The interaction between demethylation and population significantly affected leaf mass and total mass (Table 1). Demethylated plants showed a reduction in leaf mass, and this effect was only detected in the São Carlos and Piracicaba populations (native range) (Fig. 2b). On the other hand, plants from the São Carlos population (native range) reduced total biomass due to demethylation, but this trend was not found in either the Piracicaba population (native range) or the Fisterra population (non-native range) (Fig. 2e). The interaction between nutrients, demethylation and population did not significantly affect any of the variables (Table 1).

DISCUSSION

Many of the worst invasive plants around the world are clonal (Yu *et al.* 2009). This could *a priori* represent a paradox, as lack of genetic diversity associated with clonal reproduction would reduce the capacity of clonal plants for adaptation to the new environment. This situation would be even more accentuated due to genetic drift, presumably associated to the

Table 1: Results of three-way ANCOVAs for the effects of nutrient level, demethylation and population of origin on number of ramets, leaf mass, stem mass, root mass, total mass and root mass ratio of the offspring generation

		No. of ramets		Leaf mass		Stem mass		Root mass		Total mass		RMR	
Effect	df	*F*	*P*	*F*	*P*	*F*	*P*	*F*	*P*	*F*	*P*	*F*	*P*
Initial size	1	1.05	0.308	1.37	0.245	0.17	0.684	0.29	0.595	0.20	0.654	3.95	0.050
Nutrients (N)	1	17.74	**<0.001**	1.57	0.213	16.57	**<0.001**	6.19	**0.015**	11.33	**0.001**	0.75	0.390
Demethylation (D)	1	2.94	0.095	0.60	**0.015**	0.02	0.769	0.13	0.358	0.11	0.588	<0.01	0.417
Population (P)	2	81.31	**<0.001**	89.72	**<0.001**	101.4	**<0.001**	1.62	0.205	70.37	**<0.001**	154.38	**<0.001**
N × D	1	0.92	0.34	0.80	0.374	0.66	0.421	11.61	**<0.001**	3.39	0.069	19.83	**<0.001**
N × P	2	10.00	**<0.001**	2.64	0.078	6.08	**0.003**	2.83	0.065	4.54	**0.013**	3.78	**0.027**
D × P	2	1.45	0.24	6.97	**0.002**	2.70	0.073	1.73	0.184	3.91	**0.024**	1.11	0.335
N × D × P	2	0.71	0.493	1.61	0.206	1.05	0.354	0.37	0.691	0.72	0.491	1.87	0.161
Error	81												

Initial size, estimated by the number of leaves, was treated as a covariate in the model. Significant effects (P < 0.05) are shown in bold. See Figure 2 for data.

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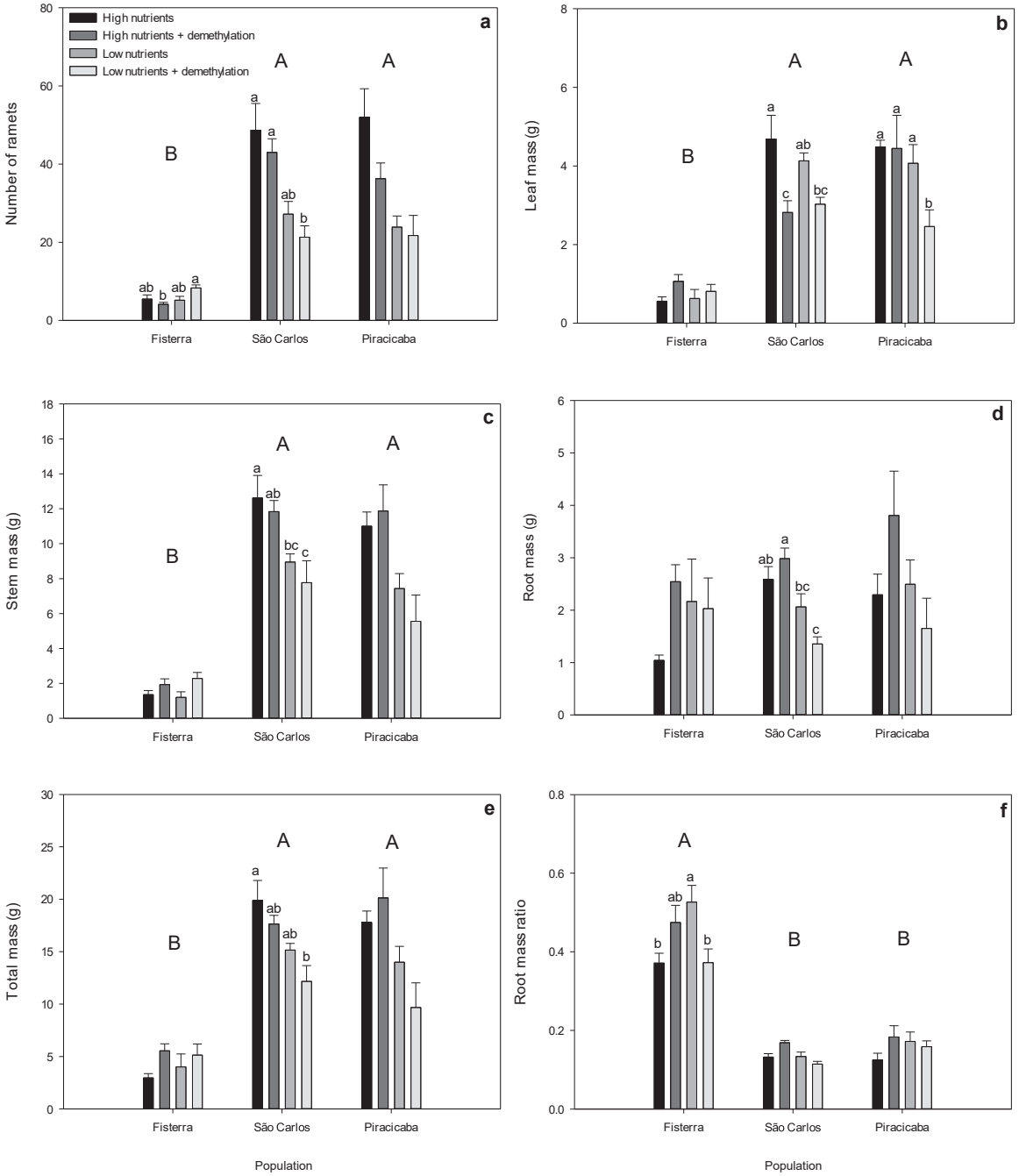

Figure 2: Number of ramets (**a**), leaf mass (**b**), stem mass (**c**), root mass (**d**), total mass (**e**) and root mass ratio (**f**) of the offspring generation derived from the three populations of *Alternanthera philoxeroides*: Fisterra (at the Iberian Peninsula, non-native range), São Carlos (at Brazil, native range) and Piracicaba (at Brazil, native range). Values are mean + SE. See Table 1 for ANOVA results. Different letters indicate differences between (upper case) and within (lower case) populations according to Tukey test ($P < 0.05$).

process of invasion, which would also reduce genetic diversity of invasive populations (Keller and Taylor 2008). In spite of this apparent limitation, clonality has been considered as a trait that could explain plant invasiveness (Pyšek 1997). In the past, many studies have been conducted to explicitly test whether traits associated with clonal growth favour the expansion of invasive plants (e.g. Chen *et al.* 2019; Roiloa *et al.* 2016; Song *et al.* 2013; Wang *et al.* 2017). However, the role of trans-generational effects in the invasiveness of clonal species has been generally overlooked (but see Dong *et al.* 2017, 2018). Our results suggest that trans-generational effects occurred in the clonal invader *A. philoxeroides*.

As predicted, environmental conditions experienced by parental generation affected the growth of the offspring generation, with ramets whose parents grew under high nutrient conditions showing greater number of ramets, stem and total mass than ramets whose parents grew in low nutrient conditions. This trans-generational effect on number of ramets, stem mass and total mass was only detected in populations from Brazil (native range), but not in the population from the Iberian Peninsula (non-native range). In addition, this trans-generational effect was not mediated by DNA methylation, indicating that other mechanisms were involved in this induced effect. One

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plausible mechanism to explain this result is that parental plants growing under high nutrient conditions would reduce the allocation of energy to produce roots, as stated by the optimal partitioning theory (Gleeson and Tilman 1992; Hilbert 1990). Therefore, plants would invest more resources in the production of above-ground structures, including the production of more or greater offspring in the case of clonal plants. In addition, the acropetal transport of resources from parent ramets established in favourable conditions will provide the offspring generation with more resources than the resources received from parents under less favourable conditions. Thus, offspring ramets that were subsided by parents growing in high nutrient conditions are expected to gain greater early advantage than those ramets supported by parents in low nutrient conditions. This initial advantage could be maintained even after disconnection, contributing to the greater total biomass shown by offspring of parent plants grown in high nutrient conditions. This transport of resources between connected units of clonal systems is mediated by physiological integration, and many previous studies have reported benefits for offspring ramets, especially when subsidized by parents growing in favourable conditions (e.g. Hartnett and Bazzaz 1983; Roiloa and Retuerto 2006; Saitoh *et al.* 2002; Slade and Hutchings 1987; Stuefer 1994). Our results fit with a 'silver-spoon' effect, where individuals developing in favourable conditions will gain performance advantages in the long term (Grafen 1988). Thus, it is expected that the environment experienced by the parental generation can contribute substantially to the phenotype of the offspring generation, and a parent plant growing in good conditions will provide more resources to their descendants (Roach and Wulff 1987). However, as mentioned above, this 'silver-spoon' effect was not detected in the population from the Iberian Peninsula (non-native range). In addition, the population from the non-native range, independent of the nutrient conditions experienced by their parental generation, performed significantly worse than populations from the native populations. A plausible explanation for these results is that much greater biomass was allocated to roots (RMR) in the non-native population in comparison with the native populations. By increasing RMR non-native plants probably reduced their net carbon gain, as the ratio between photosynthetic and non-photosynthetic structures was reduced. This would increase proportional respiration in relation with photosynthesis, and consequently reduce plant growth. The significant increase in RMR detected in non-native populations could be explained as an effect of physiological integration. Offspring ramets were disconnected from the parental plants, and consequently potential resource transport stopped. Under this situation, offspring ramets from the non-native population responded by increasing the proportional production of roots to compensate for the lack of subside received from the parental ramets. Previous experiments have demonstrated that disconnection (physiological integration impeded) conducted to a significant increase of the biomass allocated to produce roots in populations of the invasive *Carpobortus edulis* at their non-native range (Roiloa *et al.* 2013, 2019). However, to truly confirm or discard this conjecture, our study should have included a connection/disconnection treatment to determine whether there is an effect of physiological integration in biomass allocation patterns, and whether this effect differs between native and non-native populations.

Our results also showed a trans-generational effect in biomass partitioning, estimated as proportional biomass allocated to roots (RMR). Interestingly, this trans-generational effect on RMR was only detected in the population from the Iberian Peninsula (non-native range) but not in those from Brazil (native range). Furthermore, this trans-generational effect was probably mediated by DNA methylation, as the changes in biomass partitioning induced by the parental effect were not present in plants whose parental plant was subjected to demethylation. Several studies have reported the presence of epigenetic mechanism triggered by DNA methylation to explain the occurrence of trans-generational effects in plants (Bender 2004). Previous works with *A. philoxeroides* in China showed a positive correlation between phenotypic variability of plants and methylations in their DNA (Gao *et al.* 2010; Wang *et al.* 2019), although the genome of the populations was virtually identical due to the absence of sexual reproduction in the areas that the plant invades (Wang *et al.* 2005; Ye *et al.* 2003). However, although our results suggest the presence of an epigenetic control over the trans-generational effect detected in biomass partitioning, examination of the methylation patterns in the parental and offspring generation should be conducted to really elucidate the presence of a epigenetic effect modulated by DNA methylation.

Trans-generational effects on a plastic trait, such as biomass partitioning, could have important adaptive implications. Plasticity in biomass allocation allows plants to acquire resources efficiently, boosting plants to successfully colonize new environments (Mommer *et al.* 2011; Valladares *et al.* 2007). A common plastic response developed by plants is the increase of the biomass assigned to produce the structures responsible for acquiring the limiting resources, as stated by the optimal partitioning theory (Bloom *et al.* 1985; Hilbert 1990; Thornley 1972). Our results showed that offspring of parent plants grown in low nutrient conditions maintained high values of RMR (high proportion of biomass allocated to roots) even when offspring ramets grew under high nutrient conditions, denoting the presence of a trans-generational effect on biomass partitioning. Trans-generational plasticity could be considered adaptive when offspring generation gains a benefit of being early informed about their future local environment (Engqvist and Reinhold 2016). That is, when the parental environment resembles the conditions experienced by their offspring, trans-generational plasticity could represent an adaptive advantage, since offspring can anticipate their plastic response. However, when parental and offspring environments differ, benefits from trans-generational plasticity are not expected, and may even result in a penalty for offspring. Phenotypic plasticity has been suggested as an important trait contributing to plant invasiveness (Parker *et al.* 2003; Richards *et al.* 2006), and it is plausible to anticipate that phenotypic plasticity can be positively selected during the process of invasion, favouring the spread of exotic species (Lande 2015). However, the importance of adaptive trans-generational plasticity in plant invasions has received less attention. In comparison with natural selection, trans-generational plasticity would lead to a faster adaptation to local conditions, which would allow the rapid spread into the invaded environment. This could be particularly important for clonal invaders, in which lack of genetic diversity could reduce the action of natural selection, and the presence of trans-generational effects may play a key role to explain the well-known invasiveness of clonal species. Although it seems logical to predict that trans-generational effects can potentially contribute to the invasiveness of clonal species, this correlation does not necessarily always occur. Exploring in advance which traits are favouring the invasiveness of *A. philoxeroides* is mandatory to elucidate later whether trans-generational effects are operating over this trait, and consequently contributing to expansion success.

CONCLUSIONS

Our study demonstrates the existence of trans-generational effects on the invasive clonal plant *A. philoxeroides*. Trans-generational effects were observed both in populations from the native and the non-native ranges. Interestingly, trans-generational effects were detected on growth variables (number of ramets, stem mass, root mass and total mass) in the populations from the native range, and on biomass partitioning in the population from the non-native range. In addition, trans-generational effects in the population from the native range seem to be explained by a 'silver-spoon' effect. However, the effects observed in the non-native population could

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be explained by epigenetic transmission due to DNA methylation. Our results demonstrate that trans-generational effects occurred in the clonal invasive *A. philoxeroides*, and suggest that the mechanisms underlying these effects could differ between native and non-native populations. However, future studies including better representation of populations from the native and the non-native range, as well as more environments for the parental and offspring generations, must be performed to obtain a more realistic picture of the importance of trans-generational effects in the invasiveness of *A. philoxeroides*.

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